



Extrapolating a sequence

Task A: Other sequences

1) For each of the starting two terms, continue the sequence in three different ways. Explain the rule you have used for each way.

Many answers for each question, these are just some ideas.

a) 2, 6, __, __, __, ...

E.g. 2, 6, 18, 54, 162 (common multiplier of $\times 3$ between terms)

2, 6, 10, 14, 18, ... (common difference of $+4$ between terms)

2, 6, 12, 20, 30, ... (add 4, add 6, add 8, ... Common second difference $+2$)

b) -1, 5, __, __, __, ...

E.g. -1, 5, -25, 125, -625 (common multiplier of $\times -5$ between terms)

-1, 5, 4, 9, 13 (add two previous terms to get the next term)

c) 8, 3, __, __, __, ...

E.g. 8, 3, -1, -4, -6 (add -5, add -4, add -3, Common second difference $+1$)

8, 3, -2, -7, -12, ... (common difference of $+(-5)$ between terms)

d) 11, 13, __, __, __, ...

E.g. 11, 13, 15, 17, 19, ... (common difference of $+2$ between terms)

11, 13, 17, 19, 23 (Prime numbers greater than 10)

11, 13, 16, 20, 35 (Triangular numbers add 10)

2) For each of these sequences, suggest what the next two terms might be. Explain your reasoning.

There are several answers for each question. One suggestion is given for each.

a) 2, 3, 8, 17, 30, ... *Add 1, add 5, add 9, common second difference $+4$*

b) 31, 29, 31, 29, 31, ... *Sequence oscillating between 31 and 29*

c) 3, 20, 190, 1890, 18890, ... *Add 17, add 170, add 1700, difference is multiplied by 10 each time.*

d) 2, 4, 16, 256, 65536, ... *Previous term squared gives the next term*

e) 8, 5, 4, 9, 6, 5, ... *Subtract 3, subtract 1, add 5, in a repeating pattern.*

Task B: Extrapolating a sequence

1) Fill in this table to investigate patterns found when points around a circle are joined with straight lines.

Number of points on the circle	1	2	3	4	5	6
Number of lines	0	1	3	6	10	15
Number of intersections	0	0	0	1	5	13
Number of regions	1	2	4	8	16	30

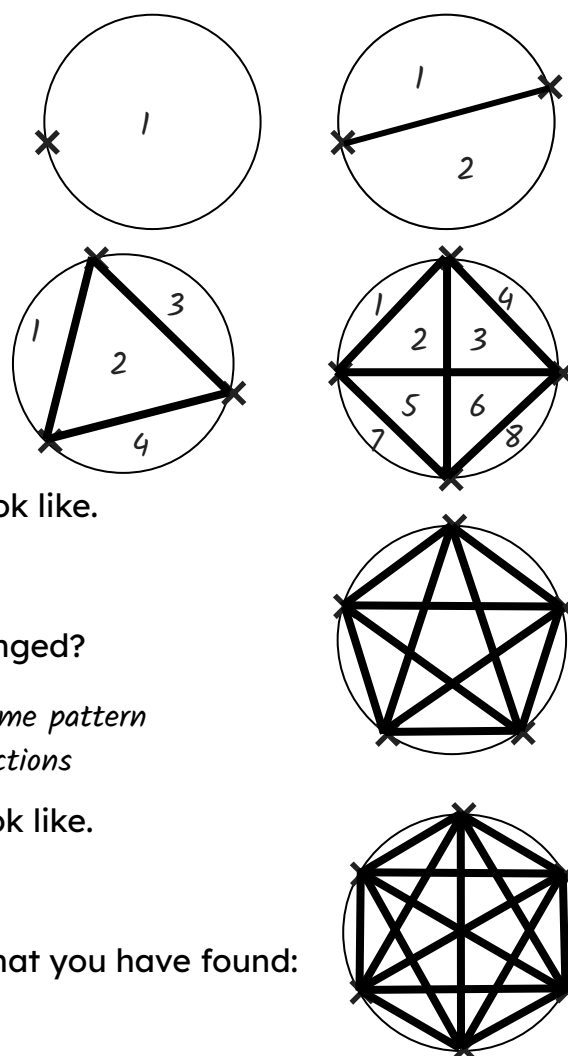
a) Complete the table for 2 points. *See table*

b) Fill in the table for 3 points and 4 points.

c) Describe any patterns you notice.

E.g. the number of lines increases by 1, then 2, then 3, For term 2 onwards these are the triangular numbers.

The number of regions are the first 4 terms in a geometric sequence.



d) Make a prediction for what the 5th column will look like.

e) Use the circle given to fill in the 5th column.

f) Were you correct, have any of your patterns changed?

*The number of lines and regions have continued in the same pattern
There are not enough terms to see a pattern with intersections*

g) Make a prediction for what the 6th column will look like.

h) Use the circle given to fill in the 6th column.

i) Were you correct? Write a sentence explaining what you have found:

Your answer may have included,

The number of regions looked geometric but this didn't continue to pattern 6. The number of lines for pattern 2 onwards is the triangular numbers. The number of intersections could have a pattern but more terms are needed.

j) Do you think you have found any patterns which will continue? Write a sentence explaining why.

The number of lines will continue to be the triangular numbers as every point is joined to every other point. The triangular number sequence is the same as the number of connections between 2, 3, 4, 5, ... objects. It is unclear how the other two patterns will continue at this stage.